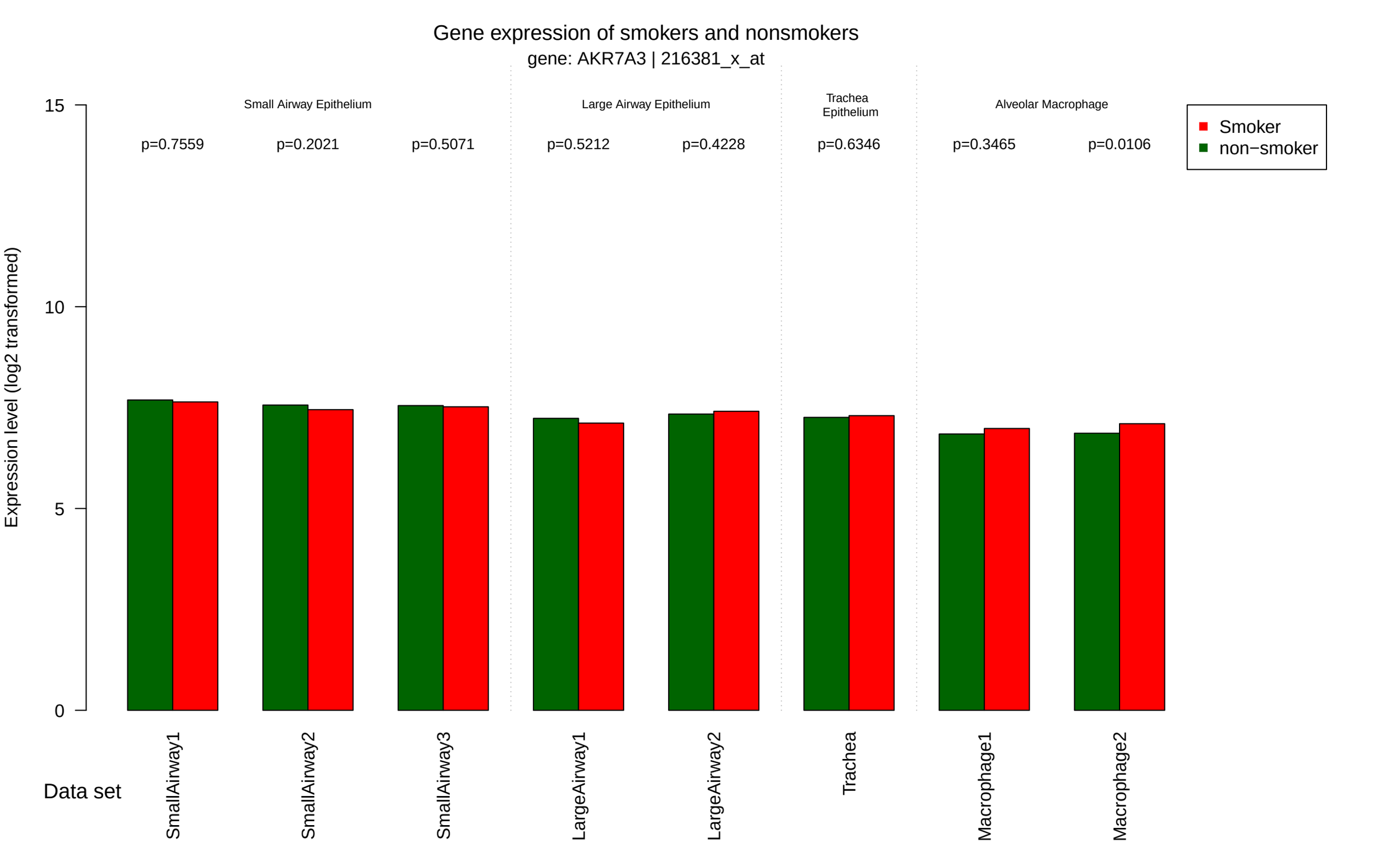
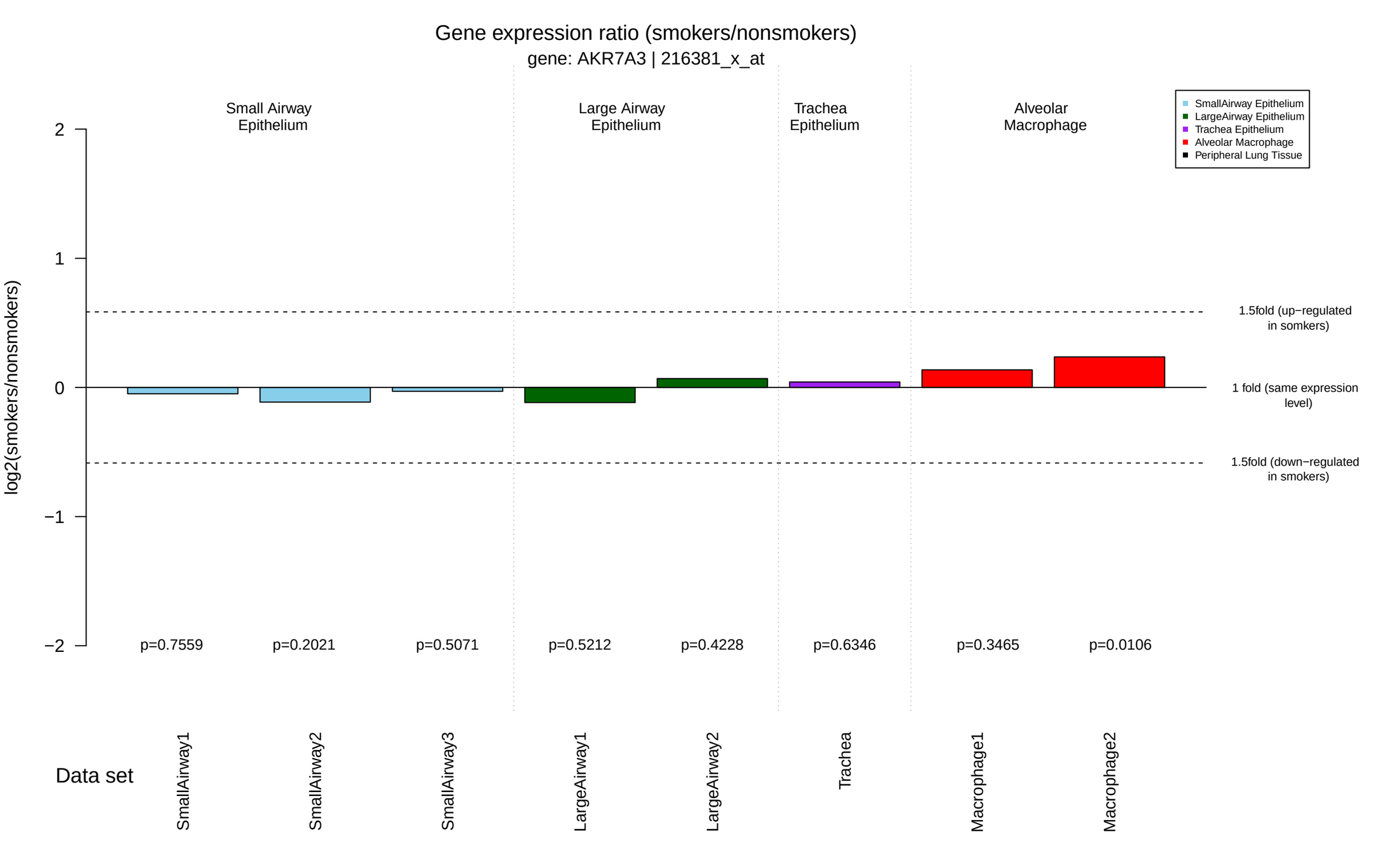
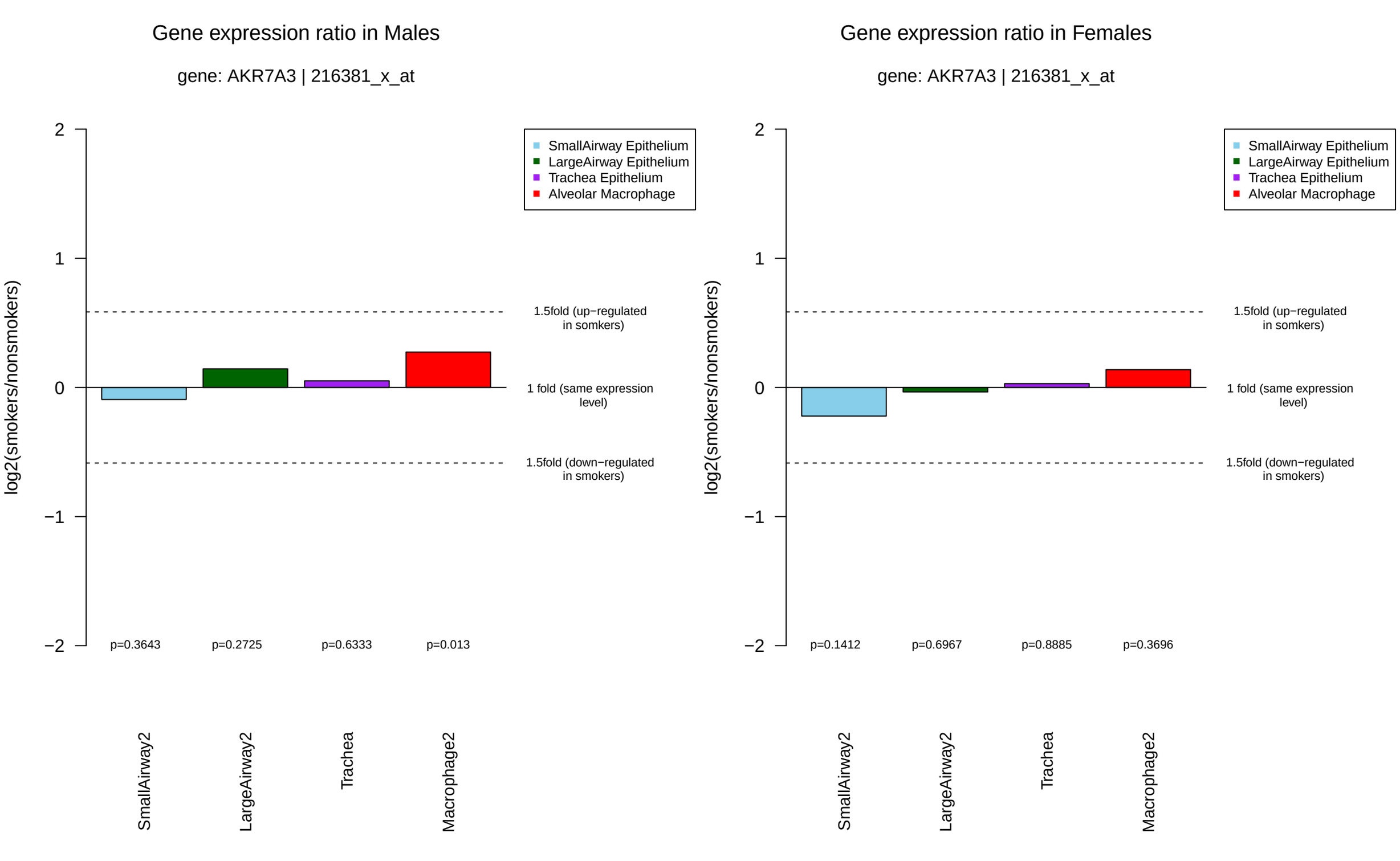
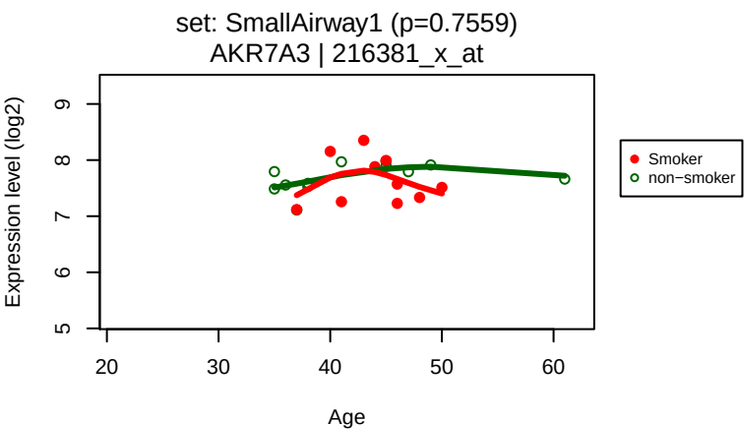
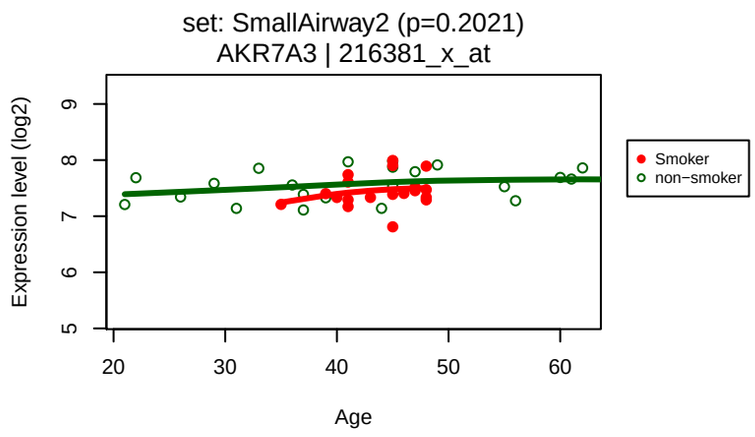
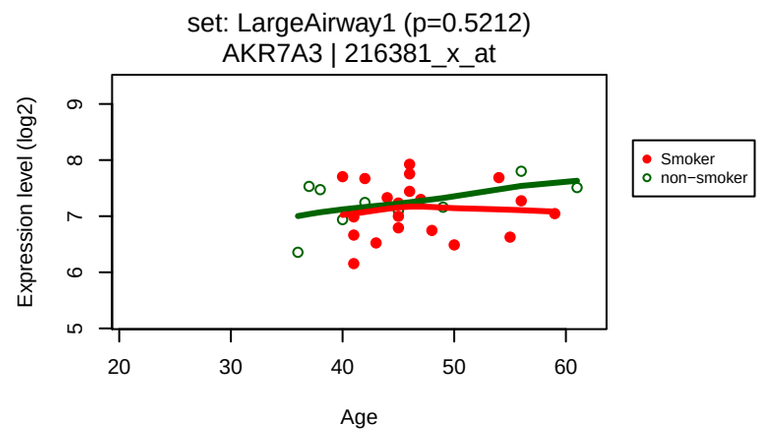

Gene expression of smokers and nonsmokers
gene: AKR7A3 | 216381_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7559
p=0.2021
p=0.5071
p=0.5212
p=0.4228
p=0.6346
p=0.3465
p=0.0106
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

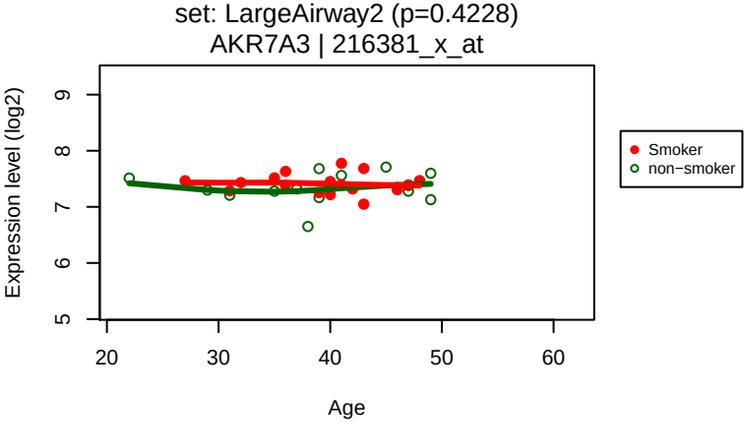

Gene expression ratio (smokers/nonsmokers)
gene: AKR7A3 | 216381_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7559
p=0.2021
p=0.5071
p=0.5212
p=0.4228
p=0.6346
p=0.3465
p=0.0106
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

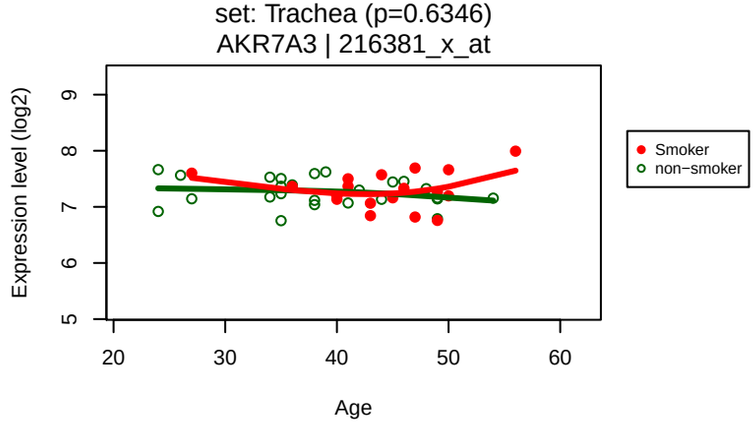

Gene expression ratio in Males
gene: AKR7A3 | 216381_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3643
p=0.2725
p=0.6333
p=0.013
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKR7A3 | 216381_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1412
p=0.6967
p=0.8885
p=0.3696
SmallAirway2
LargeAirway2
Trachea
Macrophage2

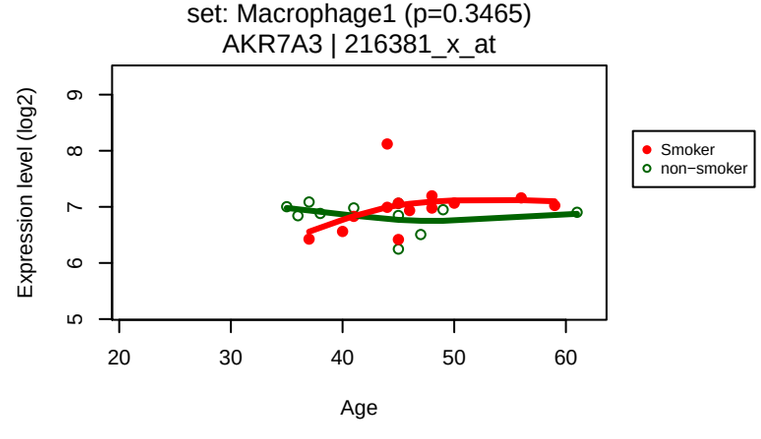

set: SmallAirway1 (p=0.7559)
AKR7A3 | 216381_x_at
Expression level (log2)
Age
Smoker
non−smoker

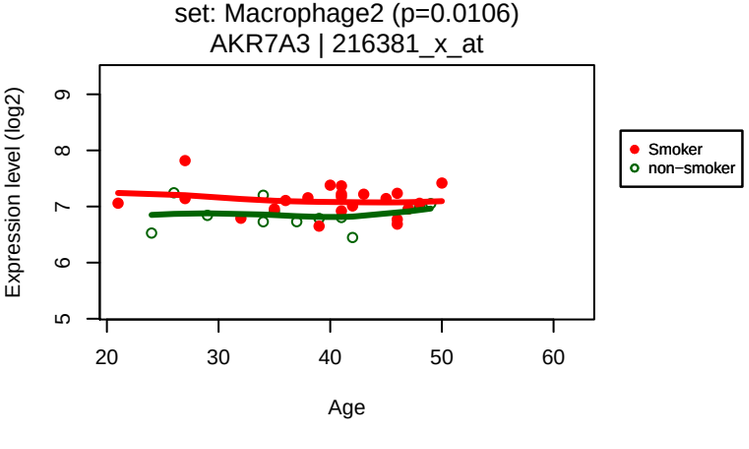

set: SmallAirway2 (p=0.2021)
AKR7A3 | 216381_x_at
Expression level (log2)
Age
Smoker
non−smoker

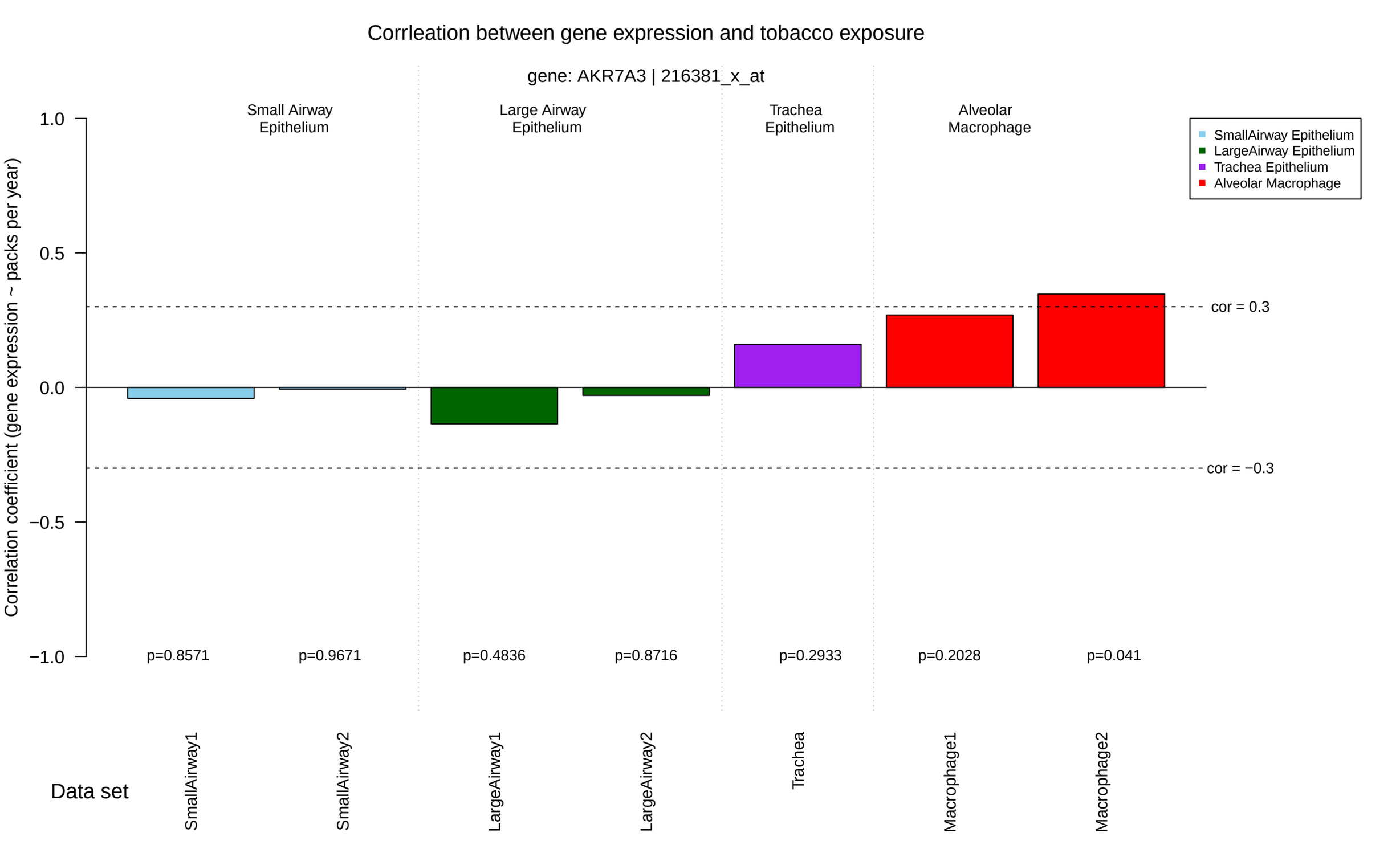

set: LargeAirway1 (p=0.5212)
AKR7A3 | 216381_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4228)
AKR7A3 | 216381_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6346)
AKR7A3 | 216381_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3465)
AKR7A3 | 216381_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0106)
AKR7A3 | 216381_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKR7A3 | 216381_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8571
p=0.9671
p=0.4836
p=0.8716
p=0.2933
p=0.2028
p=0.041
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2